R V College of Engineering
Department of Computer Science and Engineering
CIE - II: Question Paper

Subject : (Code)	**Database Management Systems (CD252IA)**	**Semester :**	**5^{TH} BE**
Date :07/01/2025	**Duration :** 120 minutes	**Staff :HR/CNS/PD/SB/SNM/PH/MNV/PT**	
Name :	**USN :**	**Section :**	**CS-A/B/C/D/CD/CY/IS/AIML**

S.N	PART-A	M	BT	Co
1.1	Consider the following tables T1 and T2 , show the result of the operation $T1 \bowtie_{(T1.P = T2.A \text{ AND } T1.R = T2.C)} T2$ T1 P Q R 10 A 5 15 B 8 25 A 6 T2 A B C 10 b 6 25 c 3 10 b 5	2	L3	4
1.2.	Write the 2 rules which must be satisfied , if a set of attributes FK in relation schema R1 is a foreign key of R1 that references relation R2	2	L1	2
1.3.	Give an example for EXISTS operator of SQL(Considering the example of Insurance Database Schema of question no. 4 of PART-B)	2	L2	3
1.4	Write reflexive, decomposition/projective inference rules for functional dependencies	2	L1	1
1.5	What is a correlated query? Give example.	2	L2	4
	PART-B			
2 a.	Explain the following relational model constraints with example. i Domain Constraint ii Key Constraint iii Entity Integrity Constraint	6	L1	2
b.	Explain how the relational model constraints that may be violated by delete operation. and the types of actions that may be taken if delete operation causes a violation	4	L2	3
3 a.	Explain the relational algebra operation for set theory with examples.	6	L1	1
b.	Explain DIVISION operation of relational algebra with an example.	4	L2	4
4	Consider the Insurance Database given below. PERSON (driver-id#:, name , address) CAR (Regno, model, year) ACCIDENT (Report-Number, date, location) OWNS (driver-id#, name, Regno) PARTICIPATED (driver-id#, Regno,Report_number, damageamount) Write the queries in relational algebra to: i Find driver-id# of every person, who owns a 'Toyota Fortuner' or a 'Hyundai Creta' car model ii Find the driver-id#, name of every person who has ever been involved some car accident iii Find the number of accidents in which cars belonging to each model were	10	L3	4

	involved iv Find the driver-id# and name of all persons who have had all of their cars involved in some accident			
5.	Consider the following Schema : Sailors(sid: integer, sname: string, rating: integer, age: real); Boats(bid: integer, bname: string, color: string); Reserves(sid: integer, bid: integer, day: date) Write the queries in SQL to: i Find the sailors information whose name begins and ends with 'A' and has at least 3 characters. ii Find the ids of sailors who have reserved a red boat or a green boat. iii Find the name of sailors who have not reserved red boat iv Find the ids and names of sailors who have reserved two different boats on the same day. v Find the average age of sailors for each rating level that has at least two sailors.	10	L3	5
6.a.	Consider the relation scheme R = {E, F, G, H, I, J, K, L, M, N} and the set of functional dependencies {{E, F} -> {G}, {F} -> {I, J}, {E, H} -> {K, L}, K -> {M}, L -> {N} on R. What is the key for R?	4	L2	3
b..	Explain with an example Aggregate functions ,Grouping and Having clause in SQL	6	L2	1

Course Outcomes:

CO1: Understand and explore the needs and concepts of relational, NoSQL database and Distributed Architecture

CO2: Apply the knowledge of logical database design principles to real time issues.

CO3: Analyze and design data base systems using relational, NoSQL and Big Data concepts

CO4: Develop applications using relational and NoSQL database

CO5: Demonstrate database applications using various technologies.

	L1	L2	L3	L4	L5	L6	CO1	CO2	CO3	CO4	CO5
Total Marks	16	22	22	-	-	-	14	08	18	10	-

R V College of Engineering
Department of Computer Science and Engineering
CIE - II: Question Paper

<table>
<tr><td>Subject : (Code)</td><td>Database Management Systems (CD252IA)</td><td colspan="2">Semester : 5TH BE</td></tr>
<tr><td>Date :07/01/2025</td><td>Duration : 120 minutes</td><td colspan="2">Staff :HR/CNS/PD/SB/SNM/PH/MNV/PT</td></tr>
<tr><td>Name :</td><td>USN :</td><td>Section :</td><td>CS-A/B/C/D/CD/CY/IS/AIML</td></tr>
</table>

<table>
<tr><th>S.N</th><th>PART-A</th><th>M</th><th>BT</th><th>Co</th></tr>
<tr><td>1.1</td><td><table><tr><td>P</td><td>Q</td><td>R</td><td>A</td><td>B</td><td>C</td></tr><tr><td>10</td><td>A</td><td>5</td><td>10</td><td>b</td><td>5</td></tr></table></td><td>2</td><td>L3</td><td>4</td></tr>
<tr><td>1.2.</td><td>1.The attributes in FK have the same domain(s) as the primary key attributes 1M PK of R2; the attributes FK are said to reference or refer to the relation R2.
2. A value of FK in a tuple t1 of the current state r1(R1) either occurs as a value of PK for some tuple t2 in the current state r2(R2) or is NULL. In the former case, we have t1[FK] = t2[PK], and we say that the tuple t1 references or refers to the tuple t2. -1M</td><td>2</td><td>L1</td><td>2</td></tr>
<tr><td>1.3.</td><td>Nested Query – Syntax 1M
Logic -1 M</td><td>2</td><td>L2</td><td>3</td></tr>
<tr><td>1.4</td><td>(reflexive rule)1: If Y ⊆ X then X->Y 1M
decomposition, or projective, rule): {X->YZ} |=X->Y -1M</td><td>2</td><td>L1</td><td>1</td></tr>
<tr><td>1.5</td><td>A correlated subquery is a subquery that contains a reference to a table that also appears in the outer query. 1M
Example -1M</td><td>2</td><td>L2</td><td>4</td></tr>
<tr><td colspan="5">PART-B</td></tr>
<tr><td>2 a.</td><td>Explain the following relational model constraints with example.
i Domain Constraint Definition and example 1.5 M
ii Key Constraint: Definition ,Example
Key ,Super Key, Candidate Key ,Primary Key 3M
iii Entity Integrity Constraint Definition and example 1,5M</td><td>6</td><td>L1</td><td>2</td></tr>
<tr><td>b.</td><td>Explain how the relational model constraints that may be violated by delete operation. and the types of actions that may be taken if delete operation causes a violation
Relational model constraints that may be violated by delete operation – Foreign Key Constraint
Example and Explanation -3 M
Action – Restrict, CASCADE, SET NULL, SET DEFAULT -1M</td><td>4</td><td>L2</td><td>3</td></tr>
<tr><td>3 a.</td><td>Explain the relational algebra operation for set theory with examples.
Union Compatibility Definition 1.5
Example and description of operator Union ,Intersection Difference -1.5 *3 =4.5</td><td>6</td><td>L1</td><td>1</td></tr>
<tr><td>b.</td><td>Scheme Definitions R,S,T - 1,5
Definition of values in T – 1.5 M</td><td>4</td><td>L2</td><td>4</td></tr>
</table>

	In general, the DIVISION operation is applied to two relations $R(Z) \div S(X)$, where the attributes of R are a subset of the attributes of S; that is, $X \subseteq Z$. Let Y be the set of attributes of R that are not attributes of S; that is, $Y = Z - X$ (and hence $Z = X \cup Y$). The result of DIVISION is a relation $T(Y)$ that includes a tuple t if tuples t_R appear in R with $t_R[Y] = t$, and with $t_R[X] = t_S$ for *every* tuple t_S in S. This means that, for a tuple t to appear in the result T of the DIVISION, the values in t must appear in R in combination with *every tuple* in S. Note that in the formulation of the DIVISION Example -1M			
4	i Find driver-id# of every person, who owns a 'Toyota Fortuner' or a 'Hyundai Creta' car model $\pi_{(driver\text{-}id\#)}(\sigma_{(model='Toyota\ Fortuner'\ or\ model='Hyundai\ Creta')}(CAR * OWNS)))$ ii Find the driver-id#, name of every person has ever been involved some car accident $\pi_{(driver\text{-}id\#)}$ (PARTICIPATED) or $\pi_{(driver\text{-}id\#,name)}$ (PARTICIPATED P $\bowtie_{(P.driver\text{-}id\# = O.\ driver\text{-}id\#)}$ OWNS O) iii Find the number of accidents in which cars belonging to each model were involved ${}_{model}\Im$count(DISTINCT Report_number) (PARTICIPATED * CAR) iv. Find the driver-id# and name of all persons who have had all of their cars involved in some accident R1<- ${}_{driver\text{-}id\#}\Im_{count(Regno)}$(OWNS) R2<- ${}_{driver\text{-}id\#}\Im_{count(DISTINCT\ Regno)}$ (PARTICIPATED) R3 <- $\pi_{(driver\text{-}id\#)}$ (R1*R2) Or Result <- $\pi_{(driver\text{-}id\#name)}$ (R3*OWNS) i	10	L3	4
5.	i Find the sailors information whose name begins and ends with 'A' and has at least 3 characters. **SELECT** * **FROM** Sailors **WHERE** name LIKE 'A_ %A' ii Find the ids of sailors who have reserved a red boat or a green boat. **SELECT DISTINCT R.sid** **FROM Boats B, Reserves R** **WHERE R.bid = B.bid AND (B.color = 'red' or B.color = 'green')** iii Find the name of sailors who have not reserved red boat **SELECT name** **FROM Sailors** **WHERE sid NOT IN** **(SELECT R.sid** **FROM Boats B, Reserves R** **WHERE R.bid = B.bid AND B.color = 'red')** iv Find the ids and names of sailors who have reserved two different boats on the same day. **SELECT DISTINCT S.sid, S.sname** **FROM Sailors S, Reserves R1, Reserves R2** **WHERE S.sid = R1.sid AND S.sid = R2.sid AND R1.day = R2.day AND R1.bid <> R2.bid** v Find the average age of sailors for each rating level that has at least two sailors. SELECT **S.rating, AVG(S.age) AS avg_age** **FROM Sailors S** **GROUP BY S.rating** **HAVING COUNT(*) >1** i	10	L3	5
6.a.	Consider the relation scheme R = {E, F, G, H, I, J, K, L, M, N} and the set of functional dependencies { {E, F} -> {G}, {F} -> {I, J},	4	L2	3

	{E, H} -> {K, L}, K -> {M}, L -> {N} $\{ E, F, H \}^{+}$ = {E, F, G, H, I, J, K, L, M, N} one of the keys. 2M Steps 2M			
b..	Explain with an example Aggregate functions ,Grouping and Having clause in SQL Expiation 3M Example -3M	6	L2	1

Course Outcomes:

CO1: Understand and explore the needs and concepts of relational, NoSQL database and Distributed Architecture
CO2: Apply the knowledge of logical database design principles to real time issues.
CO3: Analyze and design data base systems using relational, NoSQL and Big Data concepts
CO4: Develop applications using relational and NoSQL database
CO5: Demonstrate database applications using various technologies.

	L1	L2	L3	L4	L5	L6	CO1	CO2	CO3	CO4	CO5
Total Marks	16	22	22	-	-	-	14	08	18	10	-